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Hadar Fisher

hbfisher@mclean.harvard.edu

Harvard Medical School

Subigya Nepal

University of Virginia

Christian A. Webb

Harvard Medical School

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Personalized Early Detection of Depression Onset Using Multivariate Mobile Passive Sensing

Hadar Fisher, PhD[1,2]
Subigya Nepal, PhD[3]
Christian A. Webb, PhD[1,2]

[1] McLean Hospital, Belmont, MA, USA
[2] Department of Psychiatry, Harvard Medical School, Boston, MA, USA
[3] Department of Computer Science, University of Virginia, Charlottesville, VA, USA

Corresponding author:
Hadar Fisher, Ph.D.
Harvard Medical School & McLean Hospital
115 Mill Street, Belmont, MA 02478.
Email: hbfisher@mclean.harvard.edu
Phone: 617-564-6140
Fax: 617-855-4231

Abstract

Major depressive disorder is common, recurrent, and typically detected only after full symptom onset. Passive smartphone sensing offers a scalable approach to continuous, low-burden behavioral monitoring of depression risk, but prior work applying single-variable statistical process control (SPC) methods to passive sensing data has not yielded reliable prospective detection of depressive episodes. We examined whether multivariate SPC methods applied to passive sensing data could improve prospective detection of depressive symptom onset. We analyzed data from 82 college students (31.7% developed elevated depressive symptoms) followed for up to four years using continuous passive smartphone sensing. We applied exponentially weighted moving average (EWMA) control charts to individual EMA (self-esteem, stress, social level) and passive sensing variables (activity, sleep, location visits), a multivariate EWMA procedure, and a principal component analysis-based EWMA (PCA-EWMA) integrating the full passive sensing feature space. While univariate models performed poorly (MCC = .06–.21), PCA-EWMA showed stronger performance (MCC = .39–.42; sensitivity 81–85%), with up to 76.2% of alerts occurring before future depressive symptom onset. Variable contribution analyses revealed person-specific behavioral signatures driving alerts. These findings show that coordinated behavioral changes passively detected via smartphones can signal rising depressive symptoms.

Keywords: depression; passive sensing; statistical process control; early warning signals; ecological momentary assessment; digital phenotyping

Introduction

Major depressive disorder is a severe and disabling condition that imposes a substantial economic burden, strains healthcare systems, and drastically increases the risk of suicide (GBD 2019 Mental Disorders Collaborators, 2022). From both a clinical and public health perspective, the ability to anticipate increases in depressive symptoms before they fully manifest represents a critical opportunity for early identification and prevention. Achieving this goal, however, requires real-world monitoring approaches capable of detecting subtle shifts in risk before individuals cross clinical thresholds (Herrman et al., 2022).

One promising approach for this purpose is the Exponentially Weighted Moving Average (EWMA), a statistical process control (SPC) method originally developed for real-time monitoring in industrial settings (Montgomery, 2009). EWMA has recently been adapted for mental health applications, where it has been used to detect early warning signals of depression recurrence (Schat et al., 2023; Smit & Snippe, 2023; Snippe et al., 2023). In industrial contexts, SPC methods identify deviations from an established baseline in order to enable timely corrective action (Montgomery, 2009). For example, in semiconductor manufacturing, EWMA charts are used to monitor parameters such as temperature, pressure, or chemical concentrations, triggering alerts before small deviations accumulate into costly production failures. Analogously, in mental health, EWMA charts can be applied to intensive longitudinal data to track fluctuations in prodromal symptoms, alerting clinicians to emerging risk before a full depressive episode develops. A key advantage of this approach is that each individual serves as their own reference system, allowing for personalized detection of meaningful deviations from an idiographic (person-specific) baseline.

Building on this logic, several studies have applied EWMA-based monitoring to Ecological Momentary Assessment (EMA) data collected repeatedly over periods ranging from four to twelve months. These studies demonstrate that within-person changes in affective states can precede clinically meaningful symptom escalation (Schreuder et al., 2024; Smit & Snippe, 2023; Snippe et al., 2023). For instance, monitoring high-arousal negative affect, such as stress, irritation, and restlessness, yielded modest sensitivity, identifying 10 out of 22 individuals who later experienced a depressive episode, while maintaining high specificity by generating early warnings in only 2 out of 19 individuals who did not develop elevated symptoms (Snippe et al., 2023).

Importantly, in the majority of detected cases, warning signals emerged at least one month prior to symptom onset, highlighting the potential of EWMA-based approaches for prospective risk detection.

Despite these promising findings, the practical utility of EMA-based monitoring for long-term prevention remains limited. Forecasting depression often requires tracking individuals over extended periods, sometimes spanning years, in order to capture the gradual, heterogeneous processes through which vulnerability unfolds. Sustained EMA protocols place considerable burden on participants and are associated with declining adherence and high attrition, particularly in clinical populations (Astill Wright et al., 2025). These challenges fundamentally constrain the feasibility of long-term monitoring, especially when the clinical objective is to detect early warning signs over months or years rather than over brief observation windows.

Passive smartphone sensing offers a compelling, low-burden alternative for mental health monitoring, capable of collecting data virtually without patient effort (Webb & Fisher, 2025). Modern smartphones and wearables continuously generate rich streams of behavioral data, including indicators of physical activity, mobility, sleep, and social engagement, without requiring active user input (Shen et al., 2025). In principle, these data provide an unobtrusive window into behavioral patterns that are theoretically linked to depression (Choudhary et al., 2022).

Prior work on passive sensing and depression has largely focused on identifying behavioral patterns that differentiate individuals who are currently depressed from those who are not. These studies found that passive sensing variables that predict depression generally cluster around physical activity, mobility, communication and physiological patterns (Ren et al., 2023; Webb et al., 2025). Physical activity and mobility are consistent indicators, where reduced movement (e.g., lower step counts, movement index) and decreased location variability (e.g., staying at home, lower normalized entropy) are associated with higher depression severity (Aledavood et al., 2025; Fisher et al., 2025; Shen et al., 2025). Communication patterns also serve as significant predictors, though the directionality varies by context; for instance, longer duration of outgoing calls has been linked to greater depression severity (Aledavood et al., 2025), and increased call frequency is associated with worsening internalizing symptoms (Rodman et al., 2021), whereas shorter durations of incoming calls predict worse symptoms (Aledavood et al., 2025) and reduced use of active messaging apps predict worse symptoms (Choudhary et al., 2022). Regarding smartphone usage, higher overall screen time and passive consumption

(e.g., scrolling social media) are predictive of depression, whereas "active" smartphone behaviors, such as a higher frequency of unlocking the device, are associated with fewer symptoms (Choudhary et al., 2022; Marin-Dragu et al., 2023; Priftis & Panagiotakos, 2023). Finally, physiological markers, sleep disturbances (e.g., total sleep time, awakenings), are among the most frequently reported features distinguishing depression from healthy states (Shen et al., 2025).

However, the task of predicting *future* onset of depression is fundamentally different from identifying *concurrent* depression. Indeed, a recent longitudinal study applied Statistical Process Control (SPC) methods to forecast future depressive episodes in patients with bipolar disorder. Using a single variable a time, the researchers found that passive sensing parameters did not demonstrate robust detection of prodromal (pre-episode) phases or future affective shifts (Ludwig et al., 2025). One possible explanation for these null results is that monitoring single behavioral streams may fail to capture the multivariate patterns that distinguish clinically meaningful deterioration from normative variability (Beames et al., 2024). Whereas an active depressive episode may be associated with clear and stable behavioral patterns, the period preceding symptom escalation is likely marked by smaller, more diffuse changes. Rather than dramatic alterations in a single domain, early risk may manifest as subtle shifts across multiple behavioral systems, each of which may appear benign or ambiguous in isolation. As a result, features that perform well for concurrent detection may be poorly suited for forecasting future symptom emergence.

Addressing these limitations requires a shift toward multivariate, within-person analytical strategies that can capture coordinated change across multiple signals. A natural extension of EWMA to the multivariate setting is the Multivariate EWMA (MEWMA), which monitors changes in the joint mean vector of correlated variables. However, in high-dimensional contexts such as passive sensing, where the number of features can be large relative to the number of baseline observations, standard MEWMA procedures are prone to rank-deficiency and multicollinearity. PCA-based EWMA approaches address these challenges by projecting the data into a lower-dimensional space that preserves shared variance across variables while attenuating idiosyncratic noise in individual streams (Harrou et al., 2013, 2016). From a behavioral perspective, such methods align closely with theoretical models that conceptualize depression as a multi-system process, in which changes across activity, sleep, and social behavior co-occur and mutually reinforce one another over time.

The present study builds on these developments in three key ways. First, we leverage a uniquely long four-year longitudinal dataset with continuous passive smartphone sensing, enabling the examination of behavioral changes over timescales that are directly relevant for depression forecasting. Second, we apply EWMA-based monitoring procedures to examine whether depression onset can be prospectively forecasted using within-person monitoring of both passive sensing data and EMA-derived self-report measures. Third, we introduce a multivariate behavioral phenotyping framework that integrates multiple passive sensing streams using principal component analysis, enabling us to assess whether detecting subtle, coordinated behavioral shifts improves forecasting performance beyond univariate indicators. By combining long-term passive sensing with multivariate SPC methods, this study aims to clarify whether coordinated behavioral changes detectable in everyday life can serve as reliable early indicators of increasing depressive symptoms.

Method

Data Source and Participants

Data were drawn from the open-access College Experience Study, a longitudinal mobile sensing study conducted at Dartmouth College in the United States. The study followed undergraduate students for four years, from September 2017 through June 2022, using the StudentLife smartphone application to capture continuous passive sensing data alongside repeated self-reports of mental health and well-being (Nepal et al., 2024). The original study recruited 217 first-year undergraduate students across two rolling enrollment cohorts: Cohort 2021 (enrolled in 2017) and Cohort 2022 (enrolled in 2018). Participants were between 18 and 22 years old at enrollment. The sample was predominantly female (approximately 69%) and primarily identified as White (59%), with substantial representation from Asian students (23%) and smaller proportions from other racial and ethnic groups. Participants were followed throughout their undergraduate years, resulting in an unusually long and dense longitudinal record of behavior and self-reported mental health (Nepal et al., 2024).

Procedures

Participants installed the StudentLife application (R. Wang et al., 2014) on their personal iOS or Android smartphones. Once installed, the application ran continuously in the background, collecting passive sensor data with minimal participant burden. In addition, the app prompted participants to complete brief self-report surveys on a weekly basis. The original data collection was approved by the Committee for the Protection

of Human Subjects at Dartmouth College (IRB #24462). All participants in the original study provided informed consent prior to data collection.

Measures

Passive Smartphone Sensing

The StudentLife application continuously collected raw smartphone sensor data, which were processed by the original study team into interpretable behavioral features at the daily level. These sensing features captured multiple domains of daily functioning (see Supplementary **Table S1** for the full list of variables). Physical activity was indexed using accelerometer-derived estimates of activity states (e.g., still, cycling, in-vehicle). Mobility and spatial behavior were characterized using GPS-based measures of distance traveled, number of unique locations visited, and maximum distance from campus. Location context was further operationalized by classifying time spent and stationary behavior across semantic campus locations. Phone usage was assessed both globally, via the frequency and duration of phone unlock events, and contextually, by quantifying phone engagement within specific location categories. Sleep behavior was indexed using estimates of sleep duration, sleep onset, and wake times.

Ecological Momentary Assessments

Participants completed weekly self-report assessments delivered through the StudentLife app. Depressive and anxiety symptoms were assessed using the Patient Health Questionnaire-4 (PHQ-4), a brief and widely used screener (Löwe et al., 2010). To monitor depressive symptoms specifically, we derived a PHQ-2 score comprising the two depression items from the PHQ-4 (items assessing depressed mood and anhedonia over the past week). Throughout this paper, "PHQ-2" refers exclusively to this two-item depression subscale score derived from the PHQ-4, with scores ranging from 0 to 6; a score of 4 or higher indicated elevated depressive symptoms and was used as the Phase II onset threshold. The weekly surveys also included single-item measures assessing perceived stress and social activity levels, as well as a 3-item measure of self-esteem.

Statistical analysis

The College Experience Dataset is publicly available at https://www.kaggle.com/datasets/subigyanepal/college-experience-dataset, and the code used for the analyses is available on OSF at https://osf.io/ecsa5.

Analyses progressively moved from low- to high-dimensional monitoring. First, for both EMA and passive sensing data, we applied univariate EWMA control charts to each of three focal variables separately. We then jointly monitored these three variables using a multivariate EWMA framework to evaluate whether combining information across signals improved prospective detection relative to single-variable monitoring. For passive sensing data, we subsequently extended this approach to a higher-dimensional setting by applying a PCA-based EWMA framework to the full set of passive sensing features. In a final step, EMA variables were incorporated into the PCA-EWMA framework to examine whether integrating active (i.e., self-reported EMA) and passive data further improved detection.

Across all analyses, the analytic pipeline consisted of four components: (1) delineation of *Phase I* ("in-control") baseline and *Phase II* (prospective monitoring) periods, following established statistical process control conventions (Smit et al., 2023); (2) parameter calibration, including specification of smoothing parameters (λ) and control limits; (3) prospective Phase II monitoring using EWMA, MEWMA, or PCA–EWMA statistics to identify statistically meaningful deviations from each individual's baseline; and (4) evaluation of predictive performance and lead time, defined as the temporal interval between the first detected deviation from baseline and the subsequent observed increase in depressive symptoms. Full mathematical details for each component, including formula derivations, calibration procedures, and hyperparameter optimization, are provided in the Supplementary Methods.

Data Preprocessing

All analyses were conducted at the weekly level. EMA surveys were typically completed once per week, and when multiple responses occurred within a week, we used their aggregated weekly score. For passive sensing variables, raw daily smartphone-derived features were aggregated into weekly averages by computing the mean of each sensing variable across all available days within each calendar week (Sunday–Saturday). Weekly values were computed only when at least three days of valid sensing data were available; otherwise, the weekly value was coded as missing.

Consistent with EWMA procedure guidelines, which emphasize the need for a sufficiently long and stable Phase I to obtain reliable in-control estimates (typically ≥50 observations; see Smit et al., 2023; Snippe et al., 2023), we defined *Phase I* (baseline) as the first 60 valid weekly PHQ-2 scores (approximately one year of

observations) and *Phase II* as all subsequent weeks (at least 50). This threshold was chosen to ensure a sufficient number of stable, low-symptom observations for reliable baseline estimation while retaining enough Phase II weeks for meaningful prospective monitoring. Because Phase I is intended to reflect each participant's healthy or stable "in-control" period, participants were excluded if they lacked sufficient data or if any Phase I PHQ-2 score exceeded 4.

Exponentially Weighted Moving Average (EWMA) Control Charts

We applied EWMA control charts to monitor within-person changes in EMA variables (stress, self-esteem, and social interaction) and passive sensing indicators (activity level, sleep duration, and number of places visited). The EWMA statistic estimates a smoothed running average that emphasizes recent observations, enabling detection of gradual deviations from each participant's idiographic baseline. Control limits were calibrated empirically from each individual's Phase I data to ensure that limits reflect realistic baseline variability rather than overly narrow ARL-derived thresholds, which proved poorly calibrated for behavioral time series data (Schat et al., 2023, 2024). Full EWMA formulae, smoothing parameter selection, and calibration details are provided in Supplementary Methods (Section S1).

Multivariate EWMA

To simultaneously monitor changes across multiple correlated variables (e.g., stress, self-esteem, and social level), we employed a Multivariate Exponentially Weighted Moving Average (MEWMA) control chart. Unlike univariate methods that track single variables in isolation, the MEWMA procedure accounts for the correlation structure between variables, providing increased sensitivity to small or moderate shifts in the joint mean vector. All sensing features were standardized to z-scores before constructing the multivariate vector. The MEWMA statistic is based on Hotelling's T^2 evaluated against empirically calibrated upper control limits. Full MEWMA derivation and threshold selection details are provided in Supplementary Methods (Section S2).

PCA-EWMA

To address the challenges of monitoring many correlated passive sensing variables while avoiding rank-deficiency and multicollinearity issues (Kourti, 2005), we implemented a PCA-based EWMA (PCA-EWMA) framework. PCA was applied to Phase I data, and principal components explaining 90% of cumulative variance were retained. EWMA smoothing was then applied to the Hotelling's T^2 statistic computed in the reduced PC

space, and violations were signaled when this statistic exceeded empirically derived Phase I control limits. This approach was applied first to passive sensing variables alone, then with EMA variables incorporated. Hyperparameter optimization procedures are described in Supplementary Methods (Section S3).

Variable Contribution Analysis

To identify which specific variables drove detected early warning signals of elevated depressive symptoms, we decomposed the multivariate T^2 statistic at the time of threshold crossing back into the original variable space using T^2 decomposition approach (Mason et al., 1995). This step is essential for clinical interpretability, because the PCA-EWMA alert itself reflects a multivariate deviation, but does not directly indicate which specific behaviors changed. At each violation point (i.e., when the EWMA-smoothed T^2 statistic exceeded the control limit), we quantified each variable's proportional contribution to the overall T^2 value, accounting for the covariance structure learned during the individual's baseline (Phase I) period. Variables accounting for >=5% of the T^2 statistic were classified as dominant contributors; when no variable exceeded this threshold, the single strongest contributor was retained. This decomposition provides an interpretable, person-specific profile of the behavioral patterns that preceded symptom escalation. Full mathematical details are provided in Supplementary Methods (Section S4).

Performance Evaluation

Predictive performance was evaluated using sensitivity, specificity, positive and negative predictive values, and Matthews Correlation Coefficient (MCC), which provides a balanced index of detection accuracy under class imbalance. For true positive detections, lead time was calculated as the difference between the date of the first warning signal and the date of depressive (PHQ-2) symptom onset. Positive lead times indicate early warning (signals preceding symptom onset), while negative values indicate signals occurring after symptom onset.

Results

Of the 135 participants with sufficient longitudinal data (≥110 weeks of valid monitoring observations; ≥60 for Phase I and ≥50 for Phase II based on predictor availability), 53 were excluded because PHQ-2 scores during the Phase I baseline exceeded the predefined threshold (PHQ-2 ≥ 4), indicating that a stable low-symptom reference phase could not be established. The final sample (N = 82) was predominantly female

(68.3%), with 29.3% male, 1.2% identifying as both, and 1.2% missing gender data. In terms of race, 62.2% identified as White, 20.7% as Asian, 6.1% as more than one race, 4.9% as Black, and 4.9% as other/Hispanic, with 1.2% missing data. Among these participants, 26 (31.7%) experienced at least one episode of depressive symptoms exceeding the cutoff during Phase II. Among included participants, the number of valid Phase II monitoring observations had a mean of 101.0 weeks (median = 106.0 weeks, SD = 22.7), with a range of 53 to 135 weeks (approximately 2 years on average).

EMA Predictive Performance

As shown in **Table 1**, among the EMA variables, self-esteem showed the strongest overall performance (MCC = .26–.30) and a clear sensitivity–specificity trade-off across parameter settings. At the most liberal threshold (L = 3), decreases in self-esteem were detected in 18 of 26 participants (69% sensitivity) who developed elevated depressive symptoms, while 59% of participants who did not develop elevated symptoms were correctly identified as such (specificity). More conservative (higher) thresholds (L = 3.5–4.0) reduced sensitivity to 62% (16 of 26) and 50% (13 of 26), respectively, while increasing specificity to 70–79%. Across thresholds, NPV values of .77–.81 indicate that when no alert was generated, approximately 77–81% of participants did not develop elevated depressive symptoms, whereas PPV values of .36–.49 indicate that only about 36–49% of alerts corresponded to cases in which elevated depressive symptoms were observed, reflecting a substantial false-alarm rate.

Stress showed slightly lower performance (MCC=.11-.21) with relatively high sensitivity (77%), identifying the onset of depressive symptoms in up to 20 out of 26 participants who developed elevated symptoms. However, specificity was low (up to 50%), indicating a substantial number of false-positive alerts. Social level performed poorly across all thresholds, with negative MCC values, indicating performance below chance.

The multivariate EWMA (MEWMA), which monitors a vector of EMA-derived indicators simultaneously, did not outperform the best-performing univariate self-esteem model in terms of overall agreement (maximum MCC ≈ .21). Similar to stress, at H = 14, MEWMA detected impending symptom onset in 77% of participants who developed elevated depressive symptoms, but correctly identified only 45% of those who remained below the symptom threshold.

Lead-time analyses showed substantial heterogeneity in the timing of detected episodes (**Table S2**). For stress, self-esteem, and MEWMA, approximately 44–65% of true-positive alerts occurred before symptom onset, indicating clinically meaningful prodromal detection for a subset of participants. Importantly, among participants who generated alerts, even when the initial episode occurred before the alert, additional episodes often occurred after the alert, with 82–90% of alerts followed by a PHQ-2 score ≥ 4, underscoring the potential clinical utility of these alerts for ongoing monitoring and risk identification.

Passive Sensors Predictive Performance

Passive smartphone sensing variables showed more limited utility for detecting the onset of depressive symptoms (**Table 1**). Low activity was characterized by high sensitivity (81–85%) but very low specificity (25–32%), resulting in low overall agreement (MCC ≈ .06–.09). In contrast, sleep duration showed the opposite pattern, with higher specificity (71–77%) but low sensitivity (31–50%), yielding only modest performance (MCC ≈ .08–.21). Location visit frequency was uninformative, combining high sensitivity (92%) with extremely low specificity (7.5%) and near-zero agreement. Multivariate monitoring (MEWMA) modestly improved the sensitivity–specificity trade-off relative to single passive signals, but overall performance remained moderate (maximum MCC ≈ .20).

Similar to the EMA results, many alerts occurred after the first symptom episode; however, a substantial proportion (≈75–90%) were followed by subsequent episodes of elevated depressive symptoms (**Table S3**).

PCA-EWMA Results

Applying PCA-EWMA to passive sensing features substantially improved detection of depressive symptom onset relative to single-feature monitoring (MCC ≈ .39–.42; see **Table 2**). **Figure 1** provide illustrative PCA–EWMA monitoring trajectories for six representative participants. Sensitivity ranged from 81–85%, correctly identifying 21–22 of the 26 participants who experienced elevated depressive symptoms, with specificity between 59–64%. Negative predictive value was consistently high (NPV = .87–.89), suggesting that the absence of an alert reliably indicated continued low symptom levels, whereas positive predictive values were moderate (PPV = .47–.51), indicating that approximately half of alerts corresponded to true symptom elevations. Adding EMA to passive sensing features resulted in only minimal changes in overall classification accuracy.

PCA-EWMA detections were split between early and late alerts (≈45–50% each), with little change after adding EMA; nevertheless, ~70–76% of alerts were followed by a subsequent PHQ-2 ≥ 4 (**Table S4**).

Although alerts were generated using a multivariate PCA-EWMA framework, the approach allows decomposition of each alert into its dominant contributing features, enabling person-specific interpretation. Participants varied substantially in both the number (**Figure 2**) and type (**Figure 3**) of features driving alerts, with no single behavioral pattern characterizing all cases. Although certain contextual behaviors appeared more frequently among true positives (e.g., visiting religious places), these patterns were inconsistent across individuals, indicating that risk signals reflected person-specific behavioral shifts rather than a uniform set of features shared across participants.

Clinical Utility

To evaluate the potential clinical relevance of the SPC-based monitoring approaches, we examined the trade-off between sensitivity and specificity across parameter settings. Importantly, the clinical meaning of this trade-off depends on context: in some settings, prioritizing early detection may justify more false alarms, whereas in others, minimizing unnecessary alerts may be paramount. To remain agnostic to any single use case, we adopted a balanced criterion, defining a pragmatic target region of ≥70% sensitivity and ≥70% specificity as a general benchmark. As illustrated in the performance space figures (**Figures 4-6**), no signal consistently reached this predefined clinical threshold. However, several parameter configurations approached this region, reflecting a more favorable balance between sensitivity and specificity than alternative settings. For the passive-only PCA–EWMA model, the configuration closest to the predefined target region (≥ .70 sensitivity and ≥ .70 specificity) was obtained at $\lambda = .10$ and $H = 20$, yielding sensitivity of .69 and specificity of .66. When EMA-derived variables were incorporated into the PCA framework, the same parameter configuration ($\lambda = .10$, $H = 20$) produced a modest improvement in proximity to the target region, primarily through increased specificity (.68), while sensitivity remained unchanged (.69). Among univariate EMA signals, stress and self-esteem showed complementary profiles, with stress tending toward higher sensitivity and self-esteem toward higher specificity, together delineating the upper envelope of achievable performance despite falling short of the target region.

Discussion

This study examined whether statistical process control methods can be applied to EMA and passive smartphone sensing data to prospectively detect the onset of depressive symptoms. Three principal findings emerged. First, whereas passive sensing features monitored individually showed limited prospective utility, multivariate monitoring via PCA-EWMA substantially outperformed univariate approaches, achieving balanced detection accuracy (MCC = .39–.42) correctly identifying 81–85% of individuals who developed elevated depressive symptoms and correctly identifying 59–64% of those who did not. Second, univariate EWMA applied to EMA-derived self-esteem emerged as the strongest single-variable predictor, although its overall performance remained inferior to the multivariate PCA–EWMA approach applied to passive sensing features. Third, the behavioral signatures driving multivariate alerts were highly person-specific, with no single behavioral pattern characterizing all detected cases, suggesting that the prodromal phase of depression manifests through idiosyncratic rather than universal behavioral changes.

These findings extend existing work on SPC-based mental health monitoring in several important ways. Prior studies applying EWMA to EMA data demonstrated that within-person affective changes can precede depressive episodes by weeks (Helmich et al., 2023; Smit & Snippe, 2023; Snippe et al., 2023). However, these studies were limited to relatively short observation windows (4–12 months) and relied on intensive self-report, which imposes substantial participant burden. When daily assessments were reduced, detection accuracy declined (Snippe et al., 2023). Ludwig et al. (2025), in the only prior study applying SPC to passive sensing for prospective mood forecasting, found that single passive variables did not reliably detect prodromal phases in bipolar disorder. We replicate this null finding, individual passive sensing features showed poor detection accuracy when monitored univariately. However, by integrating multiple passive streams through principal component analysis, we showed that combining behavioral signals improves detection of increases in depressive symptoms. This finding suggests that risk for depression may be reflected in coordinated changes across behavioral systems rather than in any single domain. Identifying these distributed shifts may enhance the feasibility of scalable, low-burden real-time monitoring beyond tightly controlled research settings.

The superiority of PCA-EWMA over univariate monitoring aligns with theoretical models conceptualizing depression as a multi-system process involving interconnected changes across multiple domains, including sleep, social behavior, and activity (Fried et al., 2017). Recent cross-sectional evidence

reinforces the view that behavioral markers of psychopathology are distributed rather than domain-specific. Ringwald et al. (2025), mapping 27 smartphone sensor variables onto transdiagnostic psychopathology dimensions in 557 adults, found that each domain was associated with multiple distinct sensor features, and that many individual behaviors (e.g., reduced mobility) functioned as nonspecific markers of general distress. This is further supported within depression-specific research; Lekkas et al., (2023) demonstrated that distinct passive sensing streams exhibit highly heterogeneous and complementary capacities to detect specific, individual symptoms of depression, such as anhedonia versus somatic complaints. Although these findings are based on between-person associations, they are consistent with the idea that single behavioral indicators may be insufficient for identifying emerging depression. The prodromal period may be characterized not by dramatic alterations in a single domain but by subtle, coordinated shifts across multiple systems. Notably, adding EMA to the PCA-EWMA framework yielded only minimal improvement. Rather than indicating redundancy, this may reflect a methodological mismatch: subjective self-reports and continuous passive sensing operate on vastly different timescales and capture phenomenologically distinct constructs (Lekkas et al., 2023). Simply concatenating these into a single PCA may obscure the unique variance from each modality, as principal components will be dominated by the higher-dimensional passive sensing features. It is also possible that subjective and behavioral indicators represent partially independent systems during the prodromal phase, with cognitive-affective shifts (e.g., declining self-esteem) and behavioral-motor changes (e.g., reduced mobility, disrupted sleep) unfolding along distinct timelines for different individuals.

The finding that among EMA variables self-esteem showed the strongest univariate detection converges with prior SPC-based evidence from a clinically distinct population. Snippe et al., (2023) applying EWMA charts to intensive EMA data in formerly depressed patients discontinuing antidepressants, found that "negative thoughts about the self" was the single most accurate item for signaling depression recurrence (MCC = .41, sensitivity = 82%), outperforming stress, irritability, and positive affect indicators. The present study replicates this finding in a non-clinical sample over a substantially longer timescale (four years vs. four months) and using a different operationalization, weekly self-esteem ratings rather than multiple times a day. That the same construct emerges as the strongest predictor across both studies, despite differences in sample characteristics, assessment frequency, and monitoring duration, suggests that shifts in self-referential processing represent a

robust early marker of impending depression, consistent with cognitive models positioning negative self-schemas as a core vulnerability factor (Beck, 2008).

In contrast, perceived stress and self-reported social activity level, two constructs consistently linked to depression (LeMoult, 2020; Santini et al., 2015), demonstrated limited predictive utility. Stress showed high sensitivity but very low specificity, likely reflecting that stress variability is common among college students, especially during periods of academic pressure, social change, and the COVID-19 pandemic. Within the diathesis–stress framework, stress itself is a necessary but insufficient condition for depression onset, it is the interaction between stressor exposure and individual vulnerability that precipitates episodes (Monroe & Simons, 1991). This framework may explain the pattern of high sensitivity and low specificity, as stress elevations are common even without emerging depression, leading to a high false-alarm rate. Social activity level was the only variable yielding below-chance accuracy (negative MCC), likely reflecting heterogeneous and often paradoxical relationships between social behavior and emerging depression. Longitudinal evidence suggests the predominant causal direction runs from depressive symptoms to social isolation rather than the reverse (Luo, 2023), and individuals vary in whether they withdraw, increase social contact as coping style, or show no change during the prodromal period. This between-person heterogeneity may generate contradictory signals that cancel out at the group level. These findings collectively may imply that variables which reliably discriminate depressed from non-depressed states in cross-sectional designs do not necessarily possess the temporal specificity required for prospective early warning, a distinction that is critical for the design of monitoring systems intended to anticipate, rather than merely confirm, symptom escalation.

The lead-time analyses provide additional perspective on the clinical value of this monitoring approach. Across the best-performing models, approximately half of true-positive alerts were generated before symptom onset: 48–50% of PCA-EWMA alerts occurred prior to onset, mostly more than four weeks in advance, while EMA-based proactive alerts ranged from 44% (self-esteem) to 65% (stress). Critically, even among participants whose initial alert occurred after onset, 70–90% subsequently experienced elevated depressive symptoms, indicating that ostensibly "late" detections still captured clinically meaningful risk. This is significant given the well-documented delays in treatment seeking: epidemiological data indicate a median delay of 6–8 years between mood disorder onset and first treatment contact (P. S. Wang et al., 2005), with poor symptom

awareness substantially postponing help-seeking (Gurrieri et al., 2025). An automated monitoring system that flags behavioral deterioration, whether before or shortly after symptoms emerge, can potentially shorten the interval between onset and intervention.

A striking aspect of our findings is the heterogeneity of prodromal patterns across individuals. Variable contribution analyses revealed substantial between-person heterogeneity in the features driving multivariate alerts, with no single behavioral pattern characterizing all true-positive detections. This heterogeneity is consistent with the growing recognition that depression is not a monolithic disorder but a syndrome encompassing diverse symptom profiles and etiological pathways (Fried & Nesse, 2015). From a clinical standpoint, the ability to decompose multivariate alerts into person-specific behavioral drivers represents a distinct advantage over black-box prediction approaches, as it can inform tailored intervention strategies, for instance, identifying whether a particular individual's risk signal is driven primarily by sleep disruption, social withdrawal, or reduced physical activity.

Several limitations merit consideration. First, the sample comprised undergraduate students at a single institution, limiting generalizability to clinical populations and other demographic groups. The use of a brief self-report screening measure (PHQ-2) rather than clinical diagnosis means that detected "episodes" reflect elevations in self-reported symptom severity rather than formal depressive episodes. Second, passive sensing data were collected through a single smartphone application, and the availability and quality of specific sensing features varied across participants and over time, reflecting the real-world challenges of continuous mobile monitoring. Third, although no signal consistently reached the predefined clinical benchmark of 70% sensitivity and 70% specificity simultaneously, several configurations approached this threshold, and the 70/70 criterion is itself somewhat arbitrary, the optimal operating point depends on the clinical context and the relative costs of false positives versus missed detections. Fourth, the PCA-EWMA framework assumes that the principal component structure estimated during the baseline period remains relevant throughout the monitoring period. If behavioral patterns undergo fundamental reorganization, for example, due to a major life transition, the baseline decomposition may become misaligned with the current behavioral structure, potentially reducing sensitivity.

Despite these limitations, the present findings carry implications for clinical practice and future research. From a clinical perspective, the ability to passively detect coordinated behavioral shifts that precede

depressive symptom escalation, without requiring sustained self-report, addresses a fundamental barrier to long-term monitoring. The four-year observation window in the present study demonstrates that passive sensing data can sustain detection accuracy over timescales that are clinically relevant for prevention, not merely prediction. The person-specific nature of detected signals suggests SPC-based monitoring could serve as a foundation for adaptive early warning systems triggering targeted interventions when an individual's behavioral pattern deviates from their baseline. Emerging work has already begun translating passive sensing data into clinically actionable formats. Zhang et al., (2026) demonstrated that large language models can transform raw smartphone sensor streams into contextualized, human-readable narratives of daily behavioral patterns, a framework with direct potential for clinical decision support and just-in-time adaptive interventions. Such narrative-enhanced digital phenotyping could complement the SPC-based detection framework presented here by not only identifying *when* an individual's behavior deviates from baseline but also articulating *what* changed and *how* it relates to daily life context, information that could guide personalized intervention delivery and improve user engagement with monitoring systems.

From a methodological standpoint, the finding that multivariate integration was necessary to achieve acceptable detection accuracy reinforces the importance of moving beyond single-feature digital biomarkers toward composite behavioral phenotypes. Future research should examine whether incorporating additional data streams, such as social media activity, keystroke dynamics, or wearable physiological sensors, further improves detection (Tonon et al., in press), and whether the PCA-EWMA framework generalizes to clinical populations with more severe symptom trajectories. Prospective validation in independent samples and with clinical trials evaluating whether SPC-triggered interventions reduce episode onset, will be essential for translating these findings into actionable clinical tools.

In sum, this study provides evidence that coordinated behavioral changes detectable through passive smartphone sensing can potentially serve as early indicators of increasing depressive symptoms. By combining long-duration mobile phenotyping with multivariate statistical process control, we demonstrate that the signal for impending depression is distributed across behavioral domains and is best captured through integrative, person-specific monitoring frameworks. These findings offer a foundation for scalable, low-burden early warning systems that could support timely, personalized intervention for depression.

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AUTHOR CONTRIBUTIONS

H.F. conceived and designed the study, conducted the analyses, interpreted the results, and wrote the manuscript. C.A.W. supervised the project and contributed to study conceptualization and interpretation of the findings. S.N. provided the dataset and contributed to data acquisition. All authors reviewed, edited, and approved the final manuscript.

COMPETING INTERESTS

CW has received consulting fees from King & Spalding law firm but declares no non-financial competing interests. CW's interests were reviewed and are managed by McLean Hospital and Mass General Brigham in accordance with their conflict of interest policies. No funding from this entity was used to support the current work, and all views expressed are solely those of the authors. The other authors declare no competing financial or non-financial interests.

DATA AVAILABILITY STATEMENT

The College Experience Dataset is publicly available at https://www.kaggle.com/datasets/subigyanepal/college-experience-dataset, and the code used for the analyses is available on OSF at https://osf.io/ecsa5.

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Table 1

Accuracy of EWMA Charts in Signaling Elevated Symptoms of Depression

Signal	L	TP	FP	TN	FN	Sensitivity	Specificity	PPV	NPV	MCC
EMA										
Stress	3	20	35	21	6	.77	.37	.36	.78	.14
Stress	3.5	20	31	25	6	.77	.45	.39	.81	.21
Stress	4.0	16	28	28	10	.61	.50	.36	.74	.11
Self-esteem	3	18	23	33	8	.69	.59	.44	.81	.26
Self-esteem	3.5	16	17	39	10	.62	.70	.49	.80	.30
Self-esteem	4.0	13	12	44	13	.50	.79	.52	.77	.29
Social level	3	16	44	12	10	.61	.21	.27	.54	-.18
Social level	3.5	12	40	16	14	.46	.29	.23	.53	-.24
Social level	4	10	32	24	16	.38	.43	.24	.60	-.17
MEWMA	H=12	20	36	20	6	.77	.36	.36	.77	.13
MEWMA	H=14	20	32	25	6	.77	.45	.39	.81	.21
MEWMA	H=16	26	35	31	8	.77	.55	.42	.80	.23
Passive sensing										
Activity Stillness	3.0	23	46	10	3	.88	.13	.38	.75	.09
Activity Stillness	3.5	22	42	14	4	.85	.15	.38	.70	.06
Activity Stillness	4.0	21	38	14	4	.81	.20	.38	.69	.06
Location Visits	3.0	23	49	4	2	.92	.07	.32	.67	-0.01
Location Visits	3.5	23	49	4	2	.92	.07	.32	.67	-0.01
Location Visits	4.0	23	49	4	2	.92	.07	.32	.67	-0.01
Sleep Duration	L=3.0	15	23	33	11	.58	.59	.40	.75	0.16
Sleep Duration	L=3.5	13	16	40	13	.50	.71	.45	.75	0.21
Sleep Duration	L=4.0	8	13	43	18	.31	.77	.38	.70	0.08
MEWMA	H=80	23	40	13	2	.92	.24	.36	.87	0.20
MEWMA	H=100	20	34	19	5	.80	.36	.37	.79	0.16
MEWMA	H=120	18	29	24	7	.72	.45	.38	.77	0.17

Note: TP = True Positive; FP = False Positive; TN = True Negative; FN = False Negative; PPV = Positive Predictive Value; NPV = Negative Predictive Value; MCC = Matthew's Correlation Coefficient. 95% Confidence Intervals calculated using the Wilson score method. Detection threshold: PHQ-2 $\geq$ 4.

Table 2

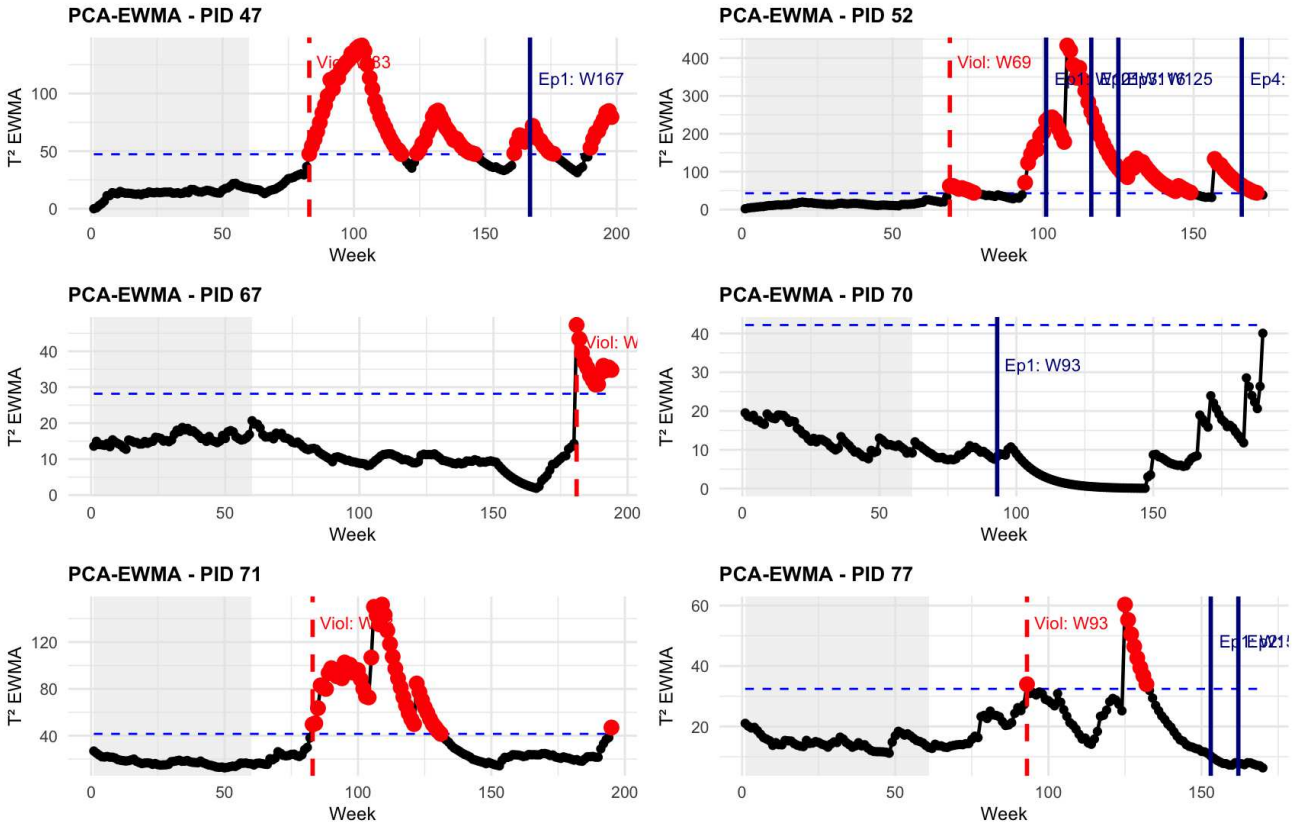
Accuracy of PCA EWMA Charts in Signaling Elevated Symptoms of Depression

Component (H)	TP	FP	TN	FN	Sensitivity	Specificity	PPV	NPV	MCC
T2 (H = 10)	22	25	31	4	.85	.55	.47	.89	.38
T2 (H = 12)	22	23	33	4	.85	.59	.49	.89	.41
T2 (H = 14)	21	22	34	5	.81	.61	.49	.87	.39
T2 (H = 16)	21	20	36	5	.81	.64	.51	.88	.42
Passive sensing + EMA									
T2 (H = 10)	23	25	31	3	.89	.55	.48	.91	.41
T2 (H = 12)	21	23	33	5	.81	.59	.48	.87	.37
T2 (H = 14)	21	21	35	5	.81	.62	.50	.88	.40
T2 (H = 16)	21	20	36	5	.81	.64	.51	.88	.42

Note. TP = True Positive; FP = False Positive; TN = True Negative; FN = False Negative; PPV = Positive Predictive Value; NPV = Negative Predictive Value; MCC = Matthews Correlation Coefficient. Detection threshold: PHQ-2 >= 4.

Figure 1

Examples of PCA-EWMA Monitoring Trajectories Across Participants



Note. Panels display weekly trajectories of the PCA-EWMA monitoring statistic (T^2) for six representative participants. Shaded regions denote the Phase I baseline period used to estimate control limits. The solid black line represents the PCA-EWMA statistic over time. The horizontal dashed blue line indicates the upper control limit. Red points denote weeks in which the PCA-EWMA statistic exceeded the control limit, indicating detected deviations from the individual's baseline. Vertical red dashed lines mark the first detected deviation. Solid blue vertical lines indicate weeks in which elevated depressive symptoms were observed.

Figure 2

Frequency Distribution of Dominant Drivers across Signal Alerts

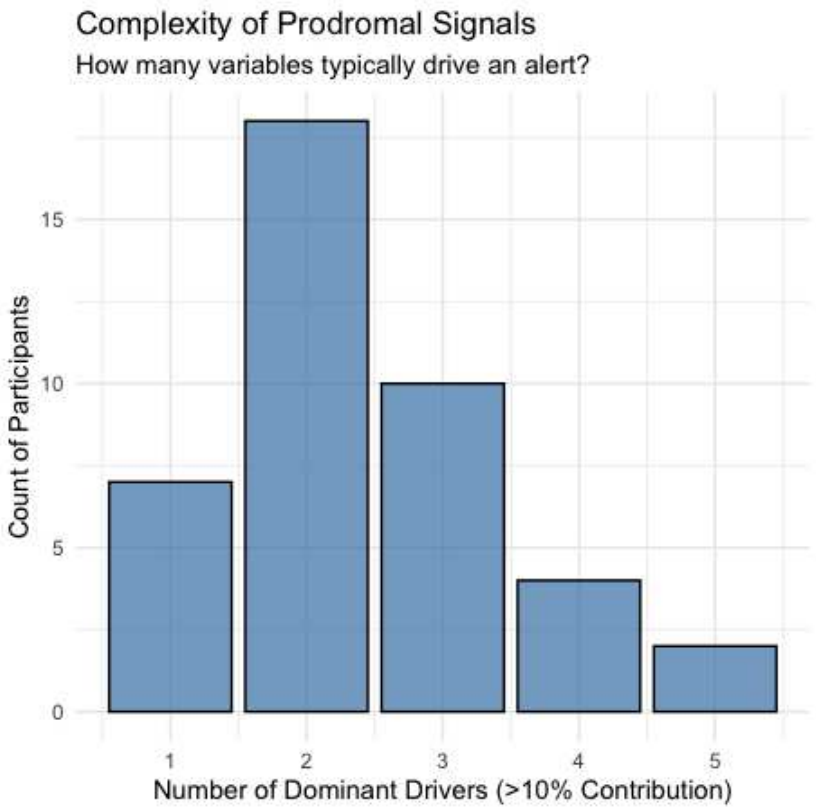


Figure 3

Passive Sensors Features Driving Alerts in True and False Positives

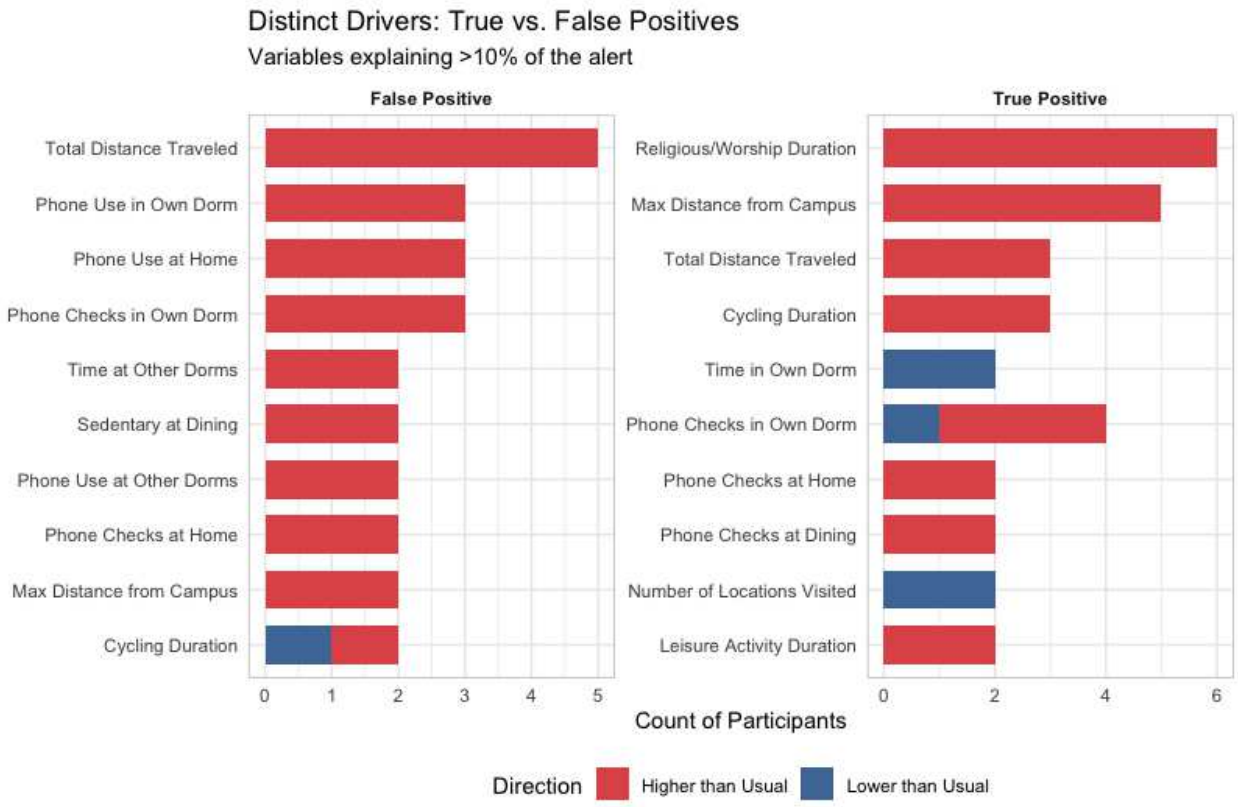


Figure 4

Sensitivity–Specificity Performance Space Across PCA-EWMA Parameter Settings (passive sensing variables)

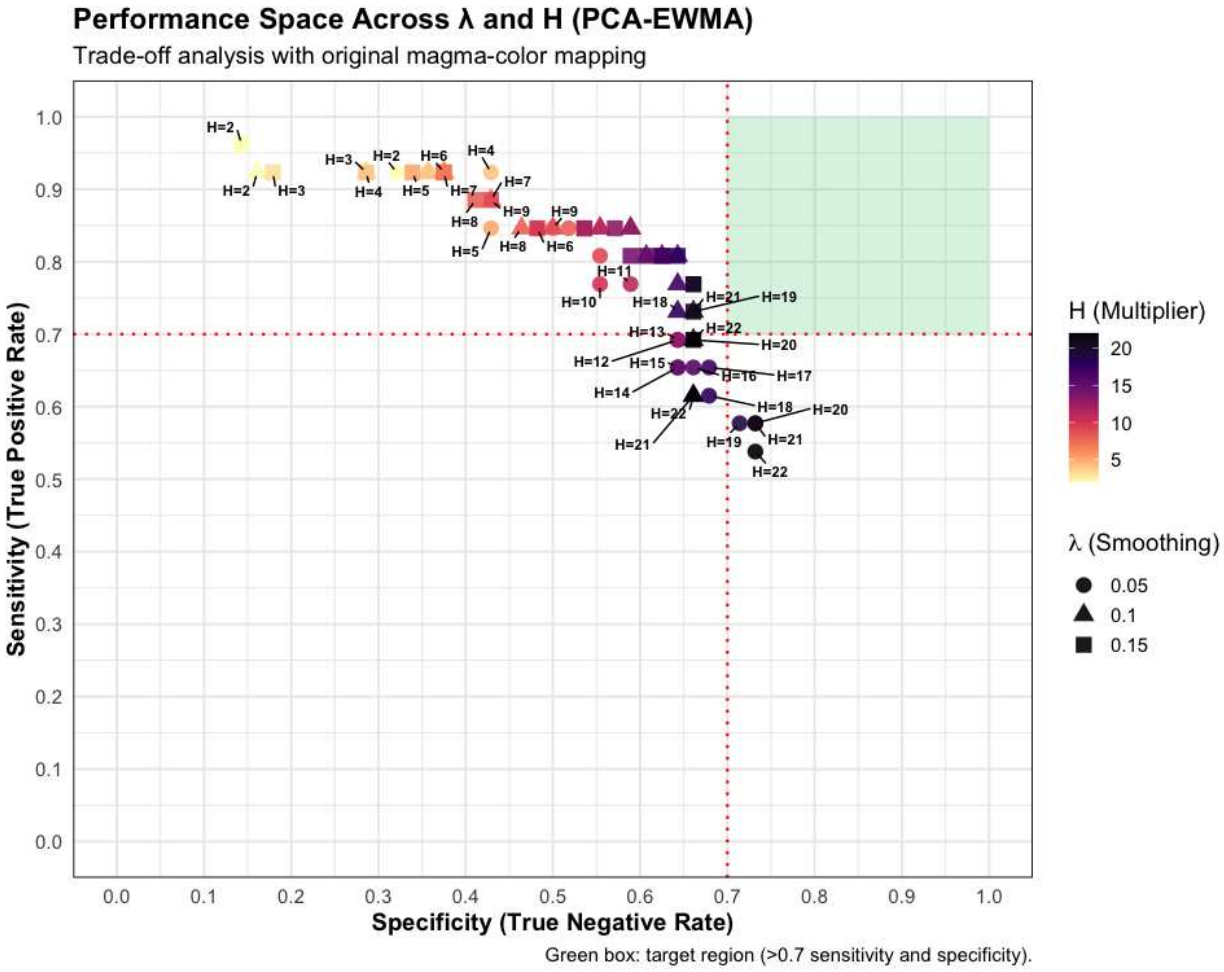


Figure 5

Sensitivity–Specificity Performance Space Across PCA-EWMA Parameter Settings (Passive Sensing + EMA Variables)

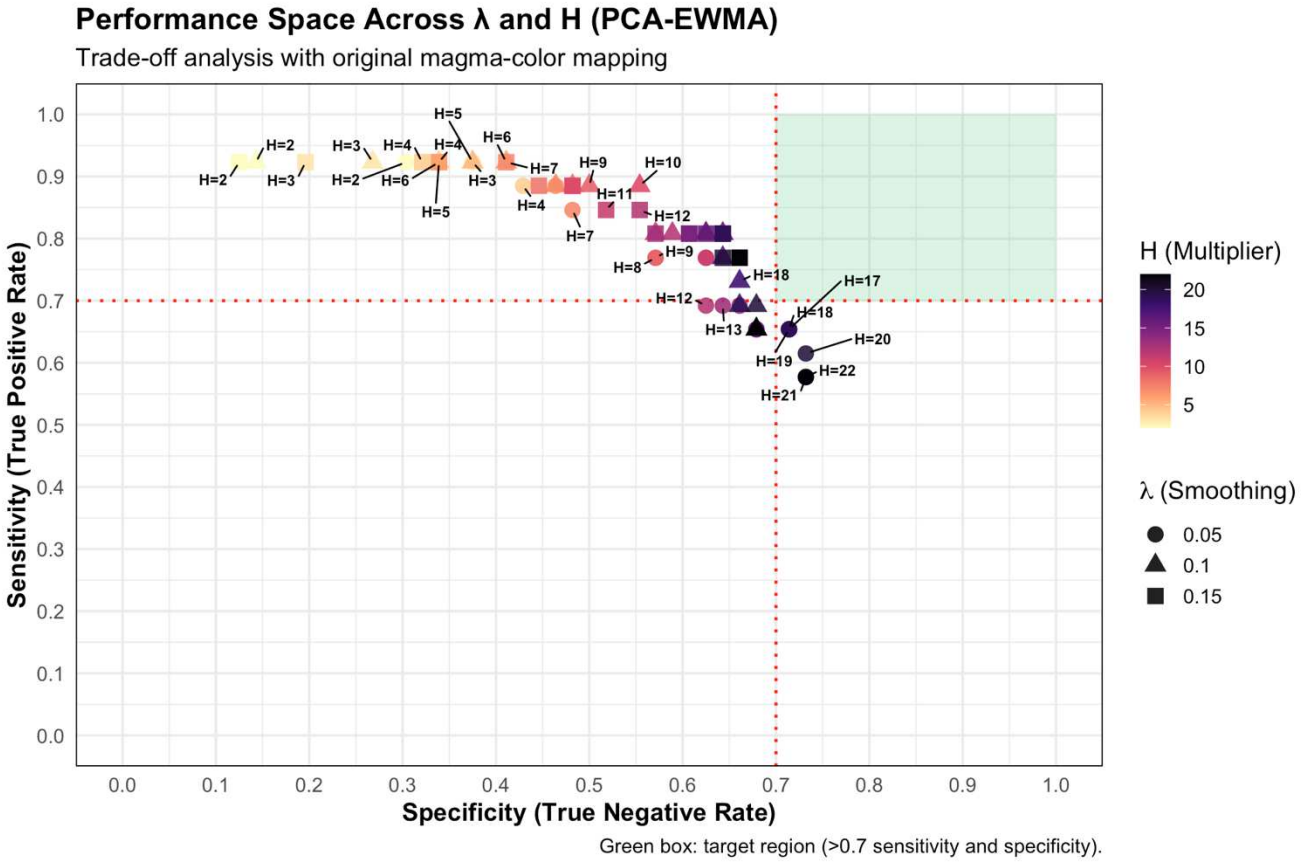
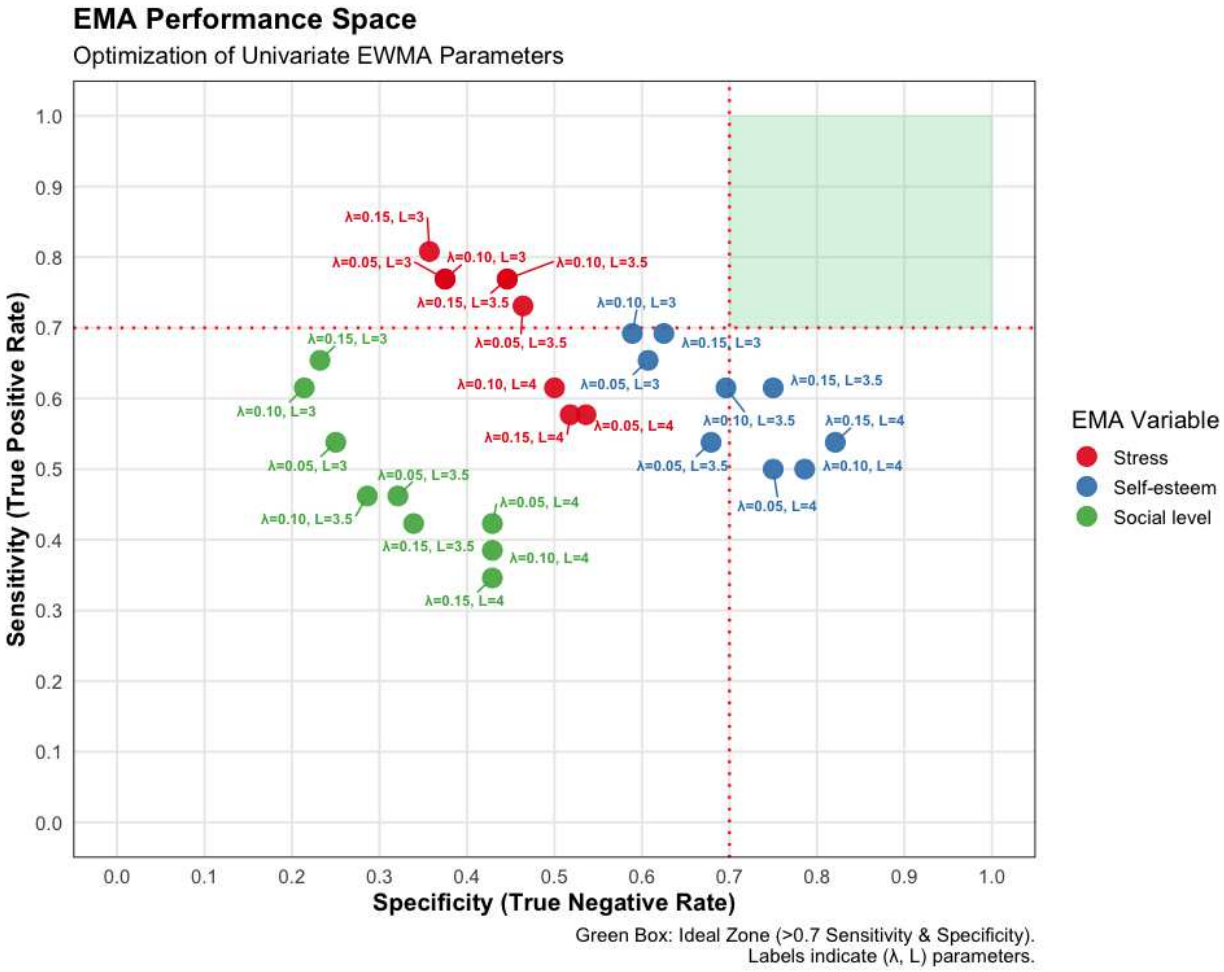


Figure 6

Sensitivity–Specificity Performance Space Across EMA-Derived EWMA Parameter Settings



Supplement

S1. Univariate EWMA: Formula, Smoothing Parameter, and Control Limit Calibration

S1.1 EWMA Formula

We first applied the Exponentially Weighted Moving Average (EWMA) procedure to monitor within-person changes in EMA variables (stress, self-esteem, and social interaction) and passive sensing indicators (activity level, sleep duration, and number of places visited). EWMA charts estimate a smoothed running average that emphasizes recent observations, allowing detection of gradual deviations from each participant's baseline. For each variable, the EWMA at week *t* was computed as:

$$\text{EWMA}_t = \lambda x_t + (1-\lambda)\text{EWMA}_{t-1},$$

where $x_t$is the weekly value of the monitored variable and λ determines the influence of the current observation. Consistent with simulation-based recommendations for EMA, we set lambda to 0.1 (Schat et al., 2023). EWMA charts rely on upper and lower control limits that characterize each participant's "in-control'' range based on Phase I data. Control limits were calculated using the individual's Phase I mean ($\hat{\mu}_1$) and standard deviation ($\hat{\sigma}_1$):

$$\text{UCL}_t = \hat{\mu}_1 + L\hat{\sigma}_1\sqrt{\frac{\lambda}{(2-\lambda)\,[1-(1-\lambda)^{2t}]}},$$

$$\text{LCL}_t = \hat{\mu}_1 - L\hat{\sigma}_1\sqrt{\frac{\lambda}{(2-\lambda)\,[1-(1-\lambda)^{2t}]}}.$$

S1.2 Control Limit Calibration

In classical SPC, the control limit width parameter (L for EWMA; H for MEWMA) is chosen to achieve a target in-control average run length (ARL_0). However, when ARL-derived values were applied to our EMA and sensing data, the resulting limits were unrealistically narrow: a substantial proportion of Phase I observations exceeded the control limits even during clinically stable periods, indicating miscalibration. Similar issues have been documented in affective time series data, where skewness, autocorrelation, and limited Phase I observations violate ARL-based assumptions, leading to inflated false alarm rates (Schat et al., 2023, 2024).

To obtain control limits reflecting realistic baseline variability, we calibrated L (and H, where applicable) empirically using only Phase I data. Specifically, we identified the smallest L such that at least 95%

of Phase I EWMA values remained within the control limits. This empirical calibration ensured that baseline variability was appropriately captured prior to Phase II monitoring.

S2. Multivariate EWMA (MEWMA): Hotelling's T² and Control Limit

For each participant at time t, the smoothed multivariate vector was computed as:

$$Z_t = \lambda X_t + (1 - \lambda) Z_{t-1}$$

where $X_t$is the vector of standardized observations for that week (e.g., stress, self-esteem, social level), λ is the smoothing constant (set to 0.10), and $Z_0$is initialized as the zero vector. Because sensing features operate on vastly different scales (e.g., minutes of sleep vs. count of locations), these variables were standardized to z-scores before constructing the multivariate vector.

To detect deviations from each participant's baseline pattern, we calculated Hotelling's T^2statistic at each time point:

$$T_t^2 = Z_t' \Sigma^{-1} Z_t$$

where Σ denotes the Phase I covariance matrix. A warning signal was issued when T_t^2 exceeded the time-varying upper control limit UCL_t, defined as:

$$UCL_t = H \times p \times \frac{\lambda}{2 - \lambda} [1 - (1 - \lambda)^{2t}]$$

where p is the number of monitored variables ($p = 3$), and H is the threshold parameter controlling the sensitivity of detection. Because the MEWMA statistic is inherently non-negative (a multivariate distance from baseline mean), only the upper control limit was used. To assess robustness, we evaluated three threshold values for H(12, 14, and 16), corresponding to increasingly conservative detection criteria.

S3. PCA-EWMA: Hotelling's T², Component Retention, and Hyperparameter Optimization

In the third phase, we implemented a Principal Component Analysis based EWMA (PCA-EWMA) framework first for only passive sensing variables and then we added EMA variables. We first applied PCA to the standardized Phase I (baseline) data and retained the set of principal components (PCs) that explained 90% of the cumulative variance. To detect deviations, for each Phase I observation, we then calculated Hotelling's T² statistic in the reduced PC space:

$$T_t^2 = s_t' \Lambda^{-1} s_t,$$

where s_t is the vector of current PC scores and Λ is the diagonal matrix of eigenvalues for the retained PCs. The T^2 statistic quantifies the squared Mahalanobis distance of observations from the Phase I centroid in the reduced PC space. Importantly, this single scalar value captures deviations in the principal-component space, detecting both extreme shifts in behavioral intensity and disruptions in the established relationships between variables. Next, we applied EWMA smoothing directly to this sequence of Phase I T^2 values:

$$Z_t = \lambda T_t^2 + (1 - \lambda) Z_{t-1},$$

From this smoothed Phase I distribution, we calculated the empirical control limits:

$$UCL = \mu_Z + H \times \sigma_Z,$$

where μ_Z and σ_Z are the mean and standard deviation of the smoothed T^2 statistic during Phase I. A violation was signaled when $Z_t > UCL$.

Hyperparameter Optimization (Q Statistic / Stability-Based Selection)

Prior to monitoring, we optimized the PCA hyperparameters (cumulative variance retained, missing data threshold, and minimum variance threshold) using a grid-search procedure. The goal was to maximize the stability of the participant-specific baseline model. Stability was quantified using the Q statistic (Squared Prediction Error), which measures the reconstruction error in the subspace orthogonal to the retained principal components:

$$Q_t = \| \mathbf{x}_t - \hat{\mathbf{x}}_t \|^2,$$

where $\mathbf{x}_t$ is the standardized observation at time t, and $\hat{\mathbf{x}}_t$ is its PCA reconstruction.

For each candidate parameter configuration, we computed the coefficient of variation of the Q statistic during Phase I:

$$CV_Q = \frac{\sigma_{Q_{\text{Phase I}}}}{\mu_{Q_{\text{Phase I}}}},$$

where $\mu_{Q_{\text{Phase I}}}$ and $\sigma_{Q_{\text{Phase I}}}$ denote the mean and standard deviation of the Q statistic across Phase I weeks. The final PCA parameters were chosen to minimize CV_Q, ensuring that the retained components formed a stable and well-captured behavioral subspace before any Phase II monitoring was conducted.

Based on this optimization procedure, we retained the number of principal components necessary to explain 90% (or the value reported by the optimizer) of the cumulative variance in Phase I.

S4. Variable Contribution Analysis (Mason-Tracy-Young T² Decomposition)

To identify which variables were responsible for elevated multivariate deviations, we decomposed the previously defined T_t^2 statistic using the Mason–Tracy–Young framework. Because $T_t^2 = s_t' \Lambda^{-1} s_t$ is computed in the retained principal component (PC) space, its value can be expressed as the sum of component-wise contributions:

$$C_{jt} = \frac{s_{jt}^2}{\lambda_j},$$

where s_{jt} denotes the score of principal component j at time t, and λ_j is the corresponding eigenvalue. By construction,

$$\sum_{j=1}^{m} C_{jt} = T_t^2.$$

To facilitate interpretation, component contributions were expressed as proportions of the total deviation:

$$c_{jt} = \frac{C_{jt}}{T_t^2}.$$

Because interpretation in principal component space is not directly clinically meaningful, contributions were projected back to the original behavioral variables using the squared PCA loadings. Let P_{kj} denote the loading of original variable k on component j. The proportional contribution of original variable k at time t was computed as:

$$C_{kt}^{\text{orig}} = \sum_{j=1}^{m} c_{jt} \, P_{kj}^2.$$

This back-projection yields a contribution profile across original passive sensing (and EMA, where applicable) variables at each alert time point, enabling identification of the behavioral features most responsible for elevated multivariate deviation. Variables accounting for ≥5% of the total contribution were classified as dominant contributors.

Table S1

Passive Smartphone Sensing Variables Included in the Analyses

Variable name	Definition
Physical activity	
act_still_ep_2	Proportion of time classified as stationary (no detected movement) based on accelerometer data (9am-6pm).
act_on_bike_ep_2	Proportion of time classified as bicycling based on accelerometer-derived activity recognition (9am-6pm).
act_in_vehicle_ep_2	Proportion of time classified as in a motor vehicle based on accelerometer-derived activity recognition (9am-6pm).
Mobility and spatial behavior	
loc_dist_ep_2	Total distance traveled during the day, derived from GPS data (9am-6pm).
loc_visit_num_ep_2	Number of unique GPS locations visited during the day (9am-6pm).
loc_max_dis_from_campus_ep_2	Maximum distance from the campus location reached during the day (9am-6pm).
Time spent in semantic location categories (duration)	
loc_food_dur	Total duration spent at food-related locations (e.g., dining halls, restaurants).
loc_health_dur	Total duration spent at health-related locations (e.g., clinics, medical facilities).
loc_home_dur	Total duration spent at home locations.
loc_leisure_dur	Total duration spent at leisure locations (e.g., entertainment, recreation).
loc_other_dorm_dur	Total duration spent in dormitories other than the participant's own.
loc_self_dorm_dur	Total duration spent in the participant's own dormitory.
loc_social_dur	Total duration spent at socially designated locations.
loc_study_dur	Total duration spent at academic or study-related locations.
loc_workout_dur	Total duration spent at workout or fitness-related locations.
loc_worship_dur	Total duration spent at places of worship.
Stationary time by location category	
loc_food_still	Duration of stationary behavior at food-related locations.
loc_home_still	Duration of stationary behavior at home locations.
loc_other_dorm_still	Duration of stationary behavior at other dormitory locations.
loc_self_dorm_still	Duration of stationary behavior at the participant's own dormitory.
loc_social_still	Duration of stationary behavior at social locations.
loc_study_still	Duration of stationary behavior at academic or study-related locations.
Context-specific phone use: unlock duration	
loc_food_unlock_duration	Total duration of phone unlock events occurring at food-related locations.
loc_home_unlock_duration	Total duration of phone unlock events occurring at home locations.
loc_other_dorm_unlock_duration	Total duration of phone unlock events occurring at other dormitories.
loc_self_dorm_unlock_duration	Total duration of phone unlock events occurring at the participant's own dormitory.
loc_social_unlock_duration	Total duration of phone unlock events occurring at social locations.

Table S1

Passive Smartphone Sensing Variables Included in the Analyses

Variable name	Definition
loc_study_unlock_duration	Total duration of phone unlock events occurring at academic or study locations.
Context-specific phone use: unlock frequency	
loc_food_unlock_num	Number of phone unlock events occurring at food-related locations.
loc_home_unlock_num	Number of phone unlock events occurring at home locations.
loc_other_dorm_unlock_num	Number of phone unlock events occurring at other dormitories.
loc_self_dorm_unlock_num	Number of phone unlock events occurring at the participant's own dormitory.
loc_social_unlock_num	Number of phone unlock events occurring at social locations.
loc_study_unlock_num	Number of phone unlock events occurring at academic or study locations.
Global phone use	
unlock_duration_ep_0	Total duration of phone unlock events during epoch (full day).
unlock_duration_ep_1	Total duration of phone unlock events during epoch 1 (12am-9am).
unlock_duration_ep_2	Total duration of phone unlock events during epoch (9am-6pm).
unlock_duration_ep_3	Total duration of phone unlock events during epoch (6pm-12am).
unlock_num_ep_0	Number of phone unlock events during epoch (full day).
unlock_num_ep_1	Number of phone unlock events during epoch (12am-9am).
unlock_num_ep_2	Number of phone unlock events during epoch (9am-6pm).
unlock_num_ep_3	Number of phone unlock events during epoch 3 (6pm-12am).
Sleep	
sleep_duration	Total sleep duration estimated from phone sensor data.
sleep_start	Estimated sleep onset time.
sleep_end	Estimated wake time.

Table S2

Lead Time (in Weeks) for Detected Depressive Episodes with ema

Combined analysis of alert timing and subsequent clinical confirmation

		Proactive Alerts (Before Onset)		Reactive Alerts (After Onset)		
	Total Alerts (N)	**Early Warning (>4w)**	**Near-term (1-4w)**	**Recent (0-4w after)**	**Late (>4w after)**	**Followed by PHQ-2 ≥ 4 (N) (%)**
Stress						
L=3	20	11 (55.0)	2 (10.0)	1 (5.0)	6 (30.0)	18 (90.0)
L=3.5	20	8 (40.0)	2 (10.0)	1 (5.0)	9 (45.0)	18 (90.0)
L=4	16	6 (37.5)	1 (6.2)	2 (12.5)	7 (43.8)	16 (81.2)
Self-esteem						
L=3	18	8 (44.4)	0 (0.0)	2 (11.1)	8 (44.4)	16 (88.9)
L=3.5	16	7 (43.8)	0 (0.0)	1 (6.2)	8 (50.0)	13 (81.2)
L=4	13	6 (46.2)	0 (0.0)	0 (0.0)	7 (53.8)	11 (84.6)
Social level						
L=3	16	5 (31.2)	0 (0.0)	1 (6.2)	10 (62.5)	12 (75.0)
L=3.5	12	4 (33.3)	0 (0.0)	0 (0.0)	8 (66.7)	8 (66.7)
L=4	10	4 (40.0)	0 (0.0)	0 (0.0)	6 (60.0)	8 (80.0)
MEWMA						
H = 12	20	10 (50.0)	1 (5.0)	0 (0.0)	9 (45.0)	18 (90.0%)
H = 14	20	10 (50.0)	1 (5.0)	0 (0.0)	9 (45.0)	18 (90.0%)
H = 16	18	8 (44.4)	0 (0.0)	2 (11.1)	8 (44.4)	16 (88.9%)

Note. "Proactive Alerts (Before Onset)" indicates alerts that preceded the first PHQ-2 >= 4 episode (row percentages within timing categories sum to 100% of true positives). "Followed by PHQ-2 >= 4" indicates alerts (whether proactive or reactive) that were followed by at least one subsequent PHQ-2 >= 4 episode during the monitoring period. A reactive alert may still be followed by an episode if additional episodes occurred after the alert.

Table S3

Lead Time (in Weeks) for Detected Depressive Episodes with Passive Sensing

	Total Alerts (N)	Proactive (Before Onset)		Reactive (After Onset)	Followed by PHQ-2 ≥ 4 (N) (%)
		Early (>4w)	Near-term (1-4w)	Late (≤0)	
Activity stillness					
L = 3.0	23	11 (47.8)	0 (0.0)	12 (52.2)	19 (82.6)
L = 3.5	22	10 (45.5)	1 (4.5)	11 (50.0)	18 (81.8)
L = 4.0	21	10 (47.6)	1 (4.8)	10 (47.6)	18 (85.7)
Location visits					
L = 3.0	23	14 (60.9)	0 (0.0)	9 (39.1)	21 (91.3)
L = 3.5	23	13 (56.5)	1 (4.3)	9 (39.1)	21 (91.3)
L = 4.0	23	12 (52.2)	1 (4.3)	10 (43.5)	21 (91.3)
Sleep duration					
L = 3.0	15	7 (46.7)	1 (6.7)	7 (46.7)	10 (66.7)
L = 3.5	13	6 (46.2)	1 (7.7)	6 (46.2)	9 (69.2)
L = 4.0	8	5 (62.5)	0 (0.0)	3 (37.5)	7 (87.5)
MEWMA					
H = 80	23	10 (43.5)	2 (8.7)	11 (47.8)	21 (91.3)
H = 100	20	6 (30.0)	1 (5.0)	13 (65.0)	17 (85.0)
H = 120	18	6 (33.3)	1 (5.6)	11 (61.1)	15 (83.3)

Note. "Proactive Alerts (Before Onset)" indicates alerts that preceded the first PHQ-2 >= 4 episode (row percentages within timing categories sum to 100% of true positives). "Followed by PHQ-2 >= 4" indicates alerts (whether proactive or reactive) that were followed by at least one subsequent PHQ-2 >= 4 episode during the monitoring period. A reactive alert may still be followed by an episode if additional episodes occurred after the alert.

Table S4

Lead Time (in Weeks) for Detected Depressive Episodes with PCA-EWMA

Multiplier (H)	Total Alerts N(%)	Proactive (Before Onset)		Reactive (After Onset)		Followed by PHQ-2 ≥ 4 (N) (%)
		Early (>4w)	Near-term (1-4w)	Recent (0-4w)	Late (>4w)	
PCA-EWMA						
10	22	11 (50.0%)	0	0	11 (50.0%)	16 (72.7%)
12	22	10 (45.5%)	0	0	12 (54.5%)	16 (72.7%)
14	21	10 (47.6%)	0	0	11 (52.4%)	16 (76.2%)
16	21	10 (47.6%)	0	0	11 (52.4%)	16 (76.2%)
PCA-EWMA with EMA						
10	23	11 (47.8%)	1 (4.3%)	0	11 (47.8%)	16 (69.6%)
12	21	11 (52.4%)	0	1 (4.8%)	9 (42.9%)	16 (76.2%)
14	21	10 (47.6%)	0	0	11 (52.4%)	16 (76.2%)
16	21	10 (47.6%)	0	0	11 (52.4%)	16 (76.2%)

Note. "Proactive Alerts (Before Onset)" indicates alerts that preceded the first PHQ-2 >= 4 episode (row percentages within timing categories sum to 100% of true positives). "Followed by PHQ-2 >= 4" indicates alerts (whether proactive or reactive) that were followed by at least one subsequent PHQ-2 >= 4 episode during the monitoring period. A reactive alert may still be followed by an episode if additional episodes occurred after the alert.